

Max

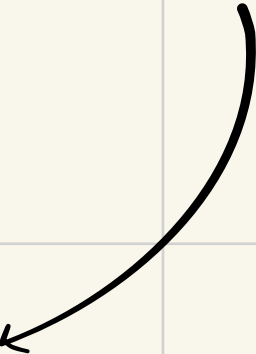


Irina



miro

pronounced "Mee-ro,"
we looked it up



Amanda



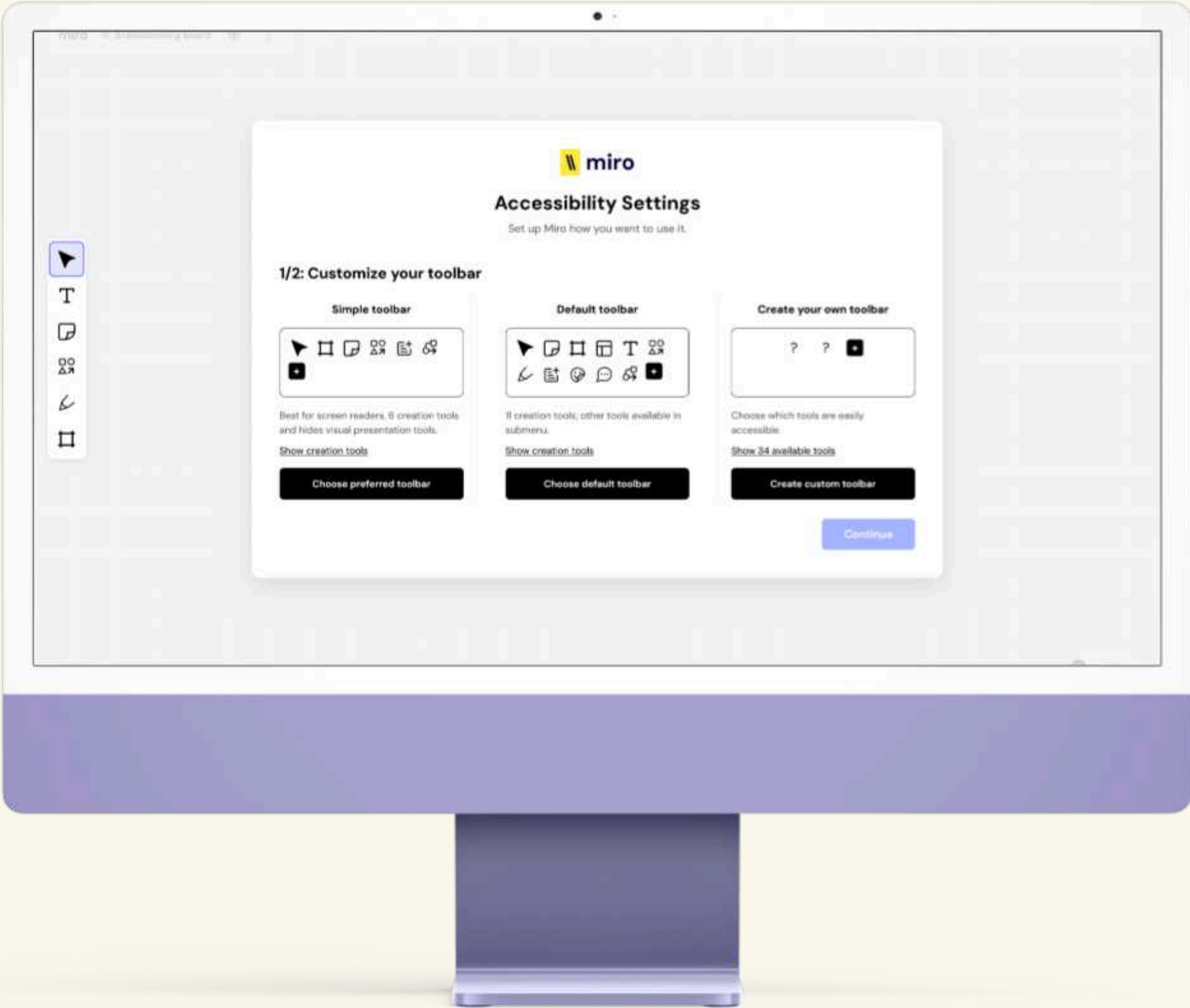
Liz



Adaptable Interfaces

Transforming Real-time Digital Whiteboard
Creation for Blind and Low Vision Users

Pathway 1: Translation



Introducing, Adaptable Interfaces.

A collection of new, customizable Miro features to assist blind and visually impaired individuals using screen readers to navigate and contribute to collaborative digital whiteboards.



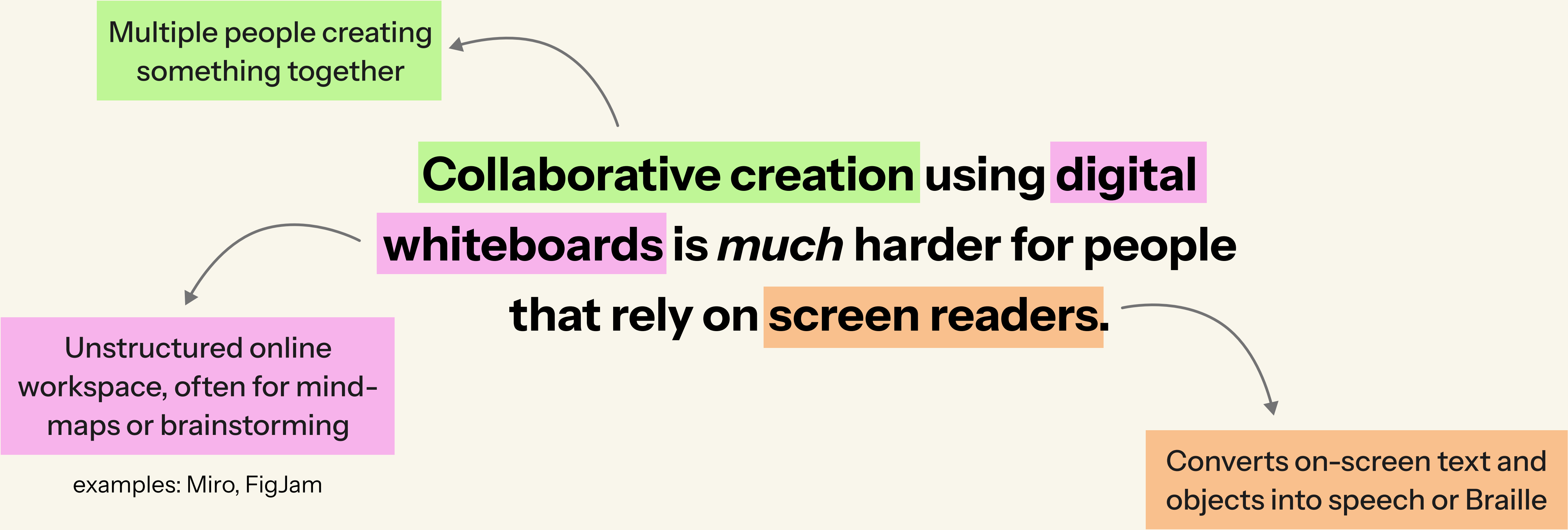
Creation tool: Miro

An online collaborative whiteboard platform. Allows teams to visually plan and brainstorm on a digital canvas.



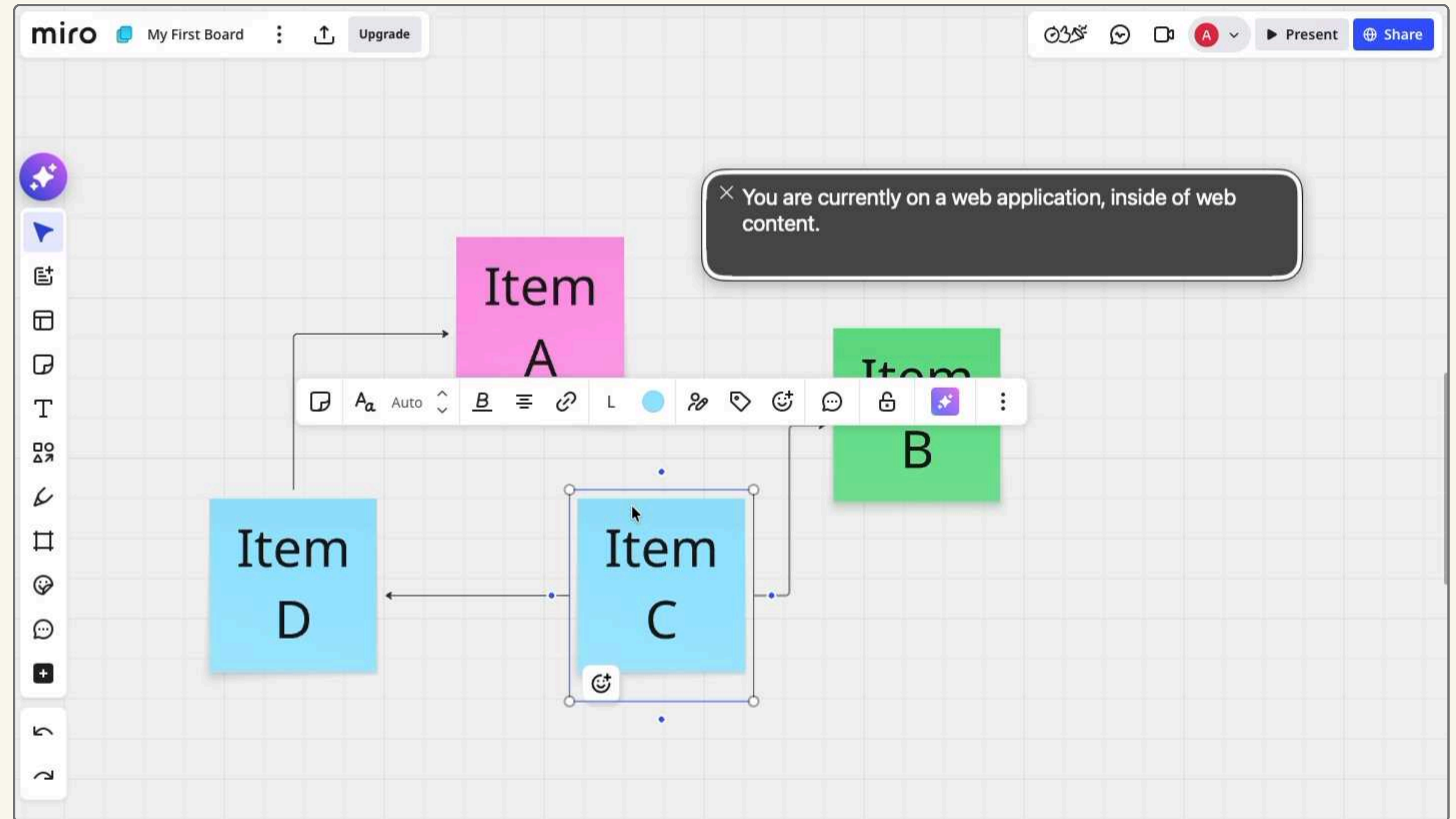
New audience: Blind/Low Vision (BLV) users

People with uncorrectable vision loss that interferes with daily activities.



Core whiteboarding features are not accessible to screen readers.

- Grouping by colour
- Grouping by proximity
- And more...



The Apple VoiceOver reader shows text and identifies the number of Connectors, but it **can't describe semantic relationships** or groups between objects

Why does this matter?

Digital whiteboards *do* make collaboration more productive and efficient. Yet, as these tools are adopted...

In schools, kids miss chances to practice brainstorming and teamwork skills.

“ [I]t is assumed students with accessibility requirements don't miss out on much. However, this is untrue as the **idea generation process** is an essential stage of learning.”

Elham Hafiz, Digital Accessibility Consultant, UT Sydney

At work, limited access to tools may impact professional development.

“ [W]hen they struggle[d] to use collaboration tools effectively, they [were] often **excluded from projects** and **unable to keep up or contribute**, leaving their colleagues frustrated and disappointed.”

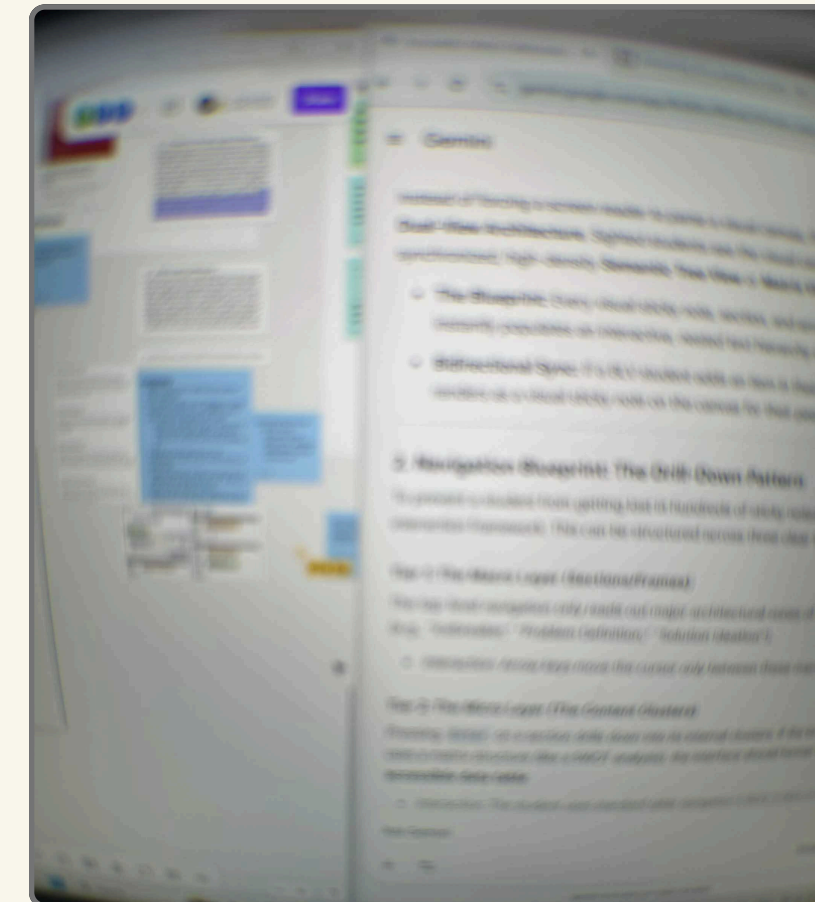
Study with 155 blind and low-vision workers

To better understand this problem space, we...

Reviewed secondary sources from experts on visual accessibility

- **Comments and articles** written by individuals with visual impairments
- **Academic studies** involving blind and visually impaired participants
- **Best practices** created by governmental and accessibility-focused organizations

Tried **simulating visual impairments** while using Miro with a screen reader*



Low vision
(removed eyeglasses)

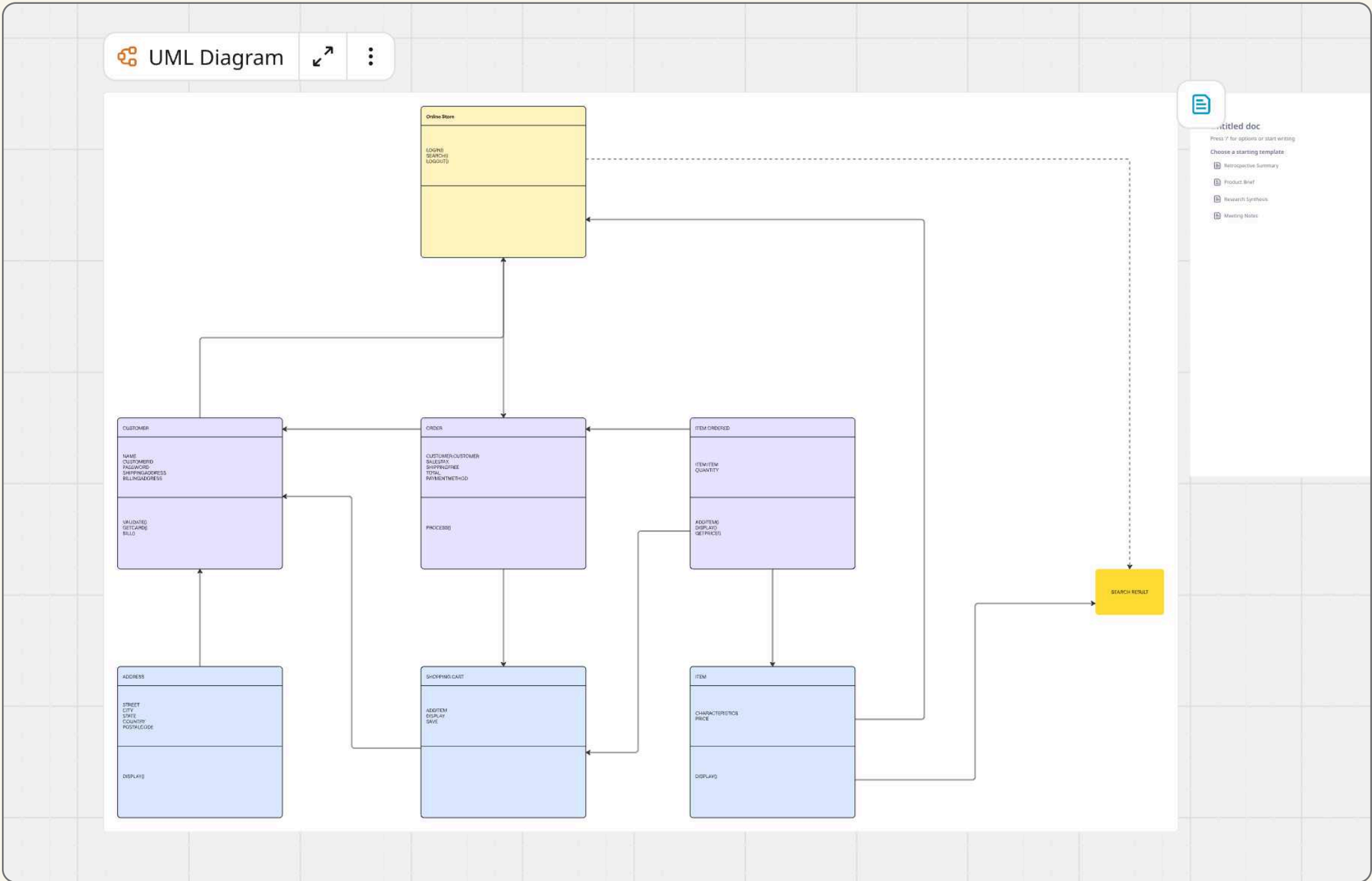


Total blindness
(covered monitor with blanket)

*As first-time screen reader users without any visual impairments, we acknowledge that our experience is limited, which affects the quality of our primary research.

Accessibility features mainly focus on navigating the *final* board, but not the **active creation process**.

Most best practices describe actions that would *later* make the whiteboard **web-accessible** (e.g. adding alt text, putting sticky notes in order).
But it’s not the same experience as seeing, understanding, and contributing in real time.



Miro diagram template: How long would it take you to understand a verbal description only?

“ [Putting screenshots into AI] is a great idea and does work well, but [it] limits you to **not having full, real time access to the data.**”

u/retrolental_morose on r/Blind

Many tools and settings aren't useful to keyboard navigating users... and instead **become distractions when looking for the useful ones.**

From zooming in to freehand drawing, several whiteboard features are great for sighted users but not relevant for others.

Over time, looking through all of them becomes **draining and time-consuming**.

“ It said 7% once and i had no idea what that meant (like i checked my laptop's battery first lol). It meant the zoom [how much I've zoomed in] ... which **doesn't mean anything** since i couldn't see.”

Comment after simulated blindness test

“ Cognitive overload is real. The screen reader does not stop talking, and navigating was hard enough. There's **no way I could've focused** on a meeting.”

Comment after simulated blindness test

“ [S]creen-reader users extract information **61.48% less accurately** and **spend 210.96% more time** interacting with online data visualizations compared to non-screen-reader users.”

Sharif et al., 2021

How might we

**How might we transform the digital
whiteboarding experience for blind and low
vision students to improve navigation?**

... it's a start.

How might we transform the digital whiteboarding experience for blind and low vision students to improve navigation?

Feedback:

Slow down. Who *are* your users, specifically? What types of *activities* are they doing on these whiteboards?

Feedback:

Remember, the project brief is focused on *creation!* Keep creation as the main experience you're focused on.

So, we went back to our research — primary *and* secondary.

Our interviewee...

- Senior on a remote team working across Canada
- Previously supervised a visually impaired employee
- Loves Miro for personal use

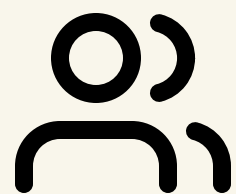
When following along with a meeting, there's too much input to effectively process all at once.

Storytime: During a Miro brainstorming activity, the visually impaired person experienced a constant stream of info that became overwhelming to process.

Others attempted to explain the board verbally, but in the end, they just had a separate 1-1 debrief with their employee.

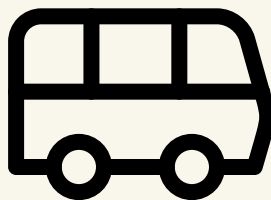
**Strengthen Our “Whys”
by Digging Deeper**

How do people accommodate blind/low-vision users in team meetings, ensuring they can collaborate effectively?



Having mediators, or teammates introducing themselves before speaking

If they are students, then surely they could meet in-person instead of having to do the creation activity online?



Most officer workers uses transportation, having to find jobs that are remote/hybrid and voiding modes of transportation



Linda

Backend Software Developer

Background

Linda works in an agile team of five on fraud systems backend. She struggles to participate in virtual scrum meetings using Miro which limits her potential to work with her teammates and consequentially stifles her professional development.

What she's tried so far

Her PM and the team try to be accommodating by adhering to Miro's best practices in accessibility and verbally communicating what they're referencing on the board.

Frustrations

- **time consuming** to write down her thoughts
- teammate assisting with writing for her
- navigating the interface is **unreliable with a screen reader**
- inaccurately positioning elements on the board

Her needs

- **easily collaborate with her team** in brainstorming session
- **comprehend teammates' contribution** to stay informed board so she can and **meaningfully contribute.**

Old HMW

How might we transform the digital whiteboarding experience for blind and low vision students to improve navigation?

New HMW

How might we make the Miro creation process more **adaptable** and **structured** for **first-time** blind and low vision users?

Adaptable

Can adjust to the needs and preferences of multiple types of blindness

Structured

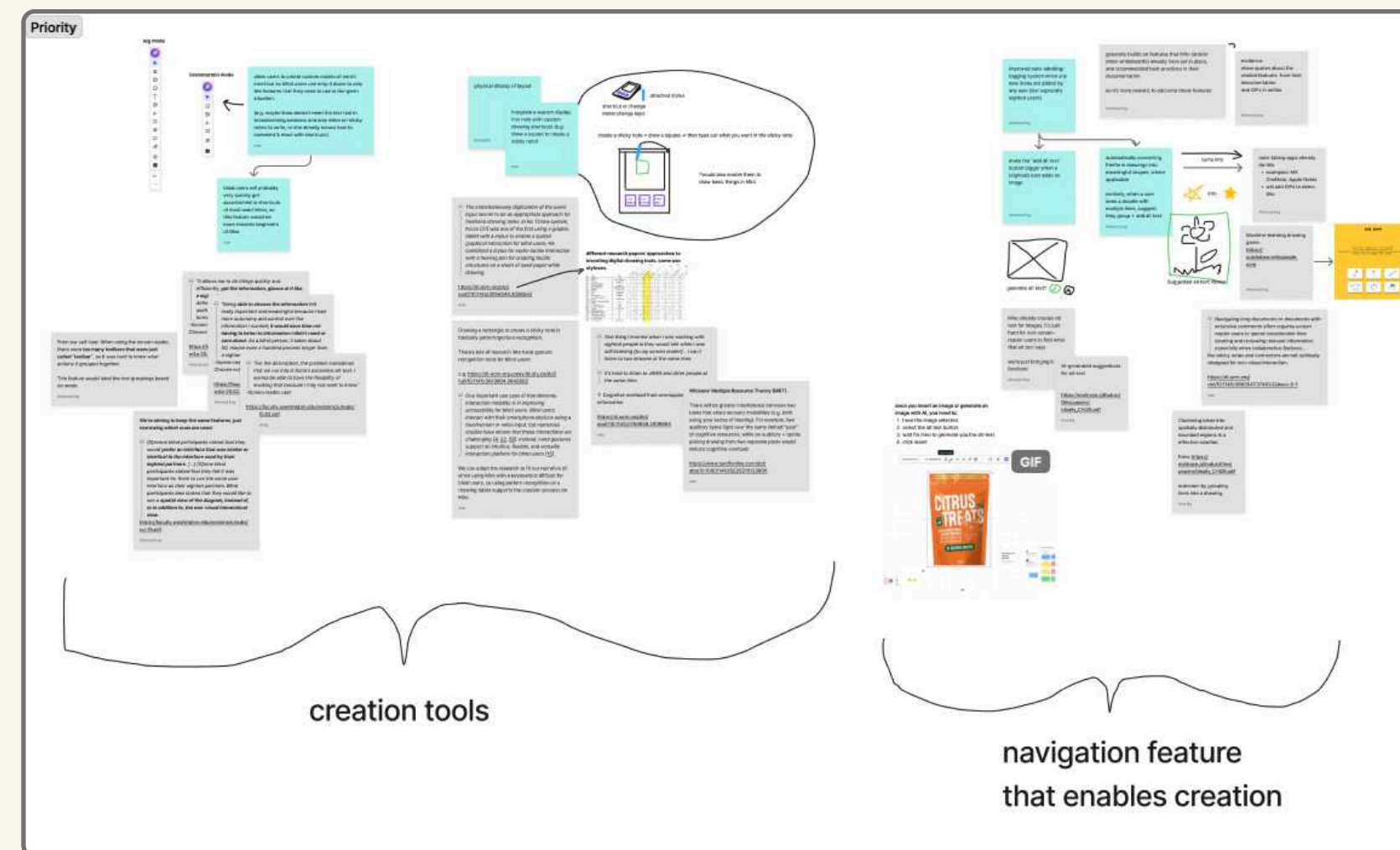
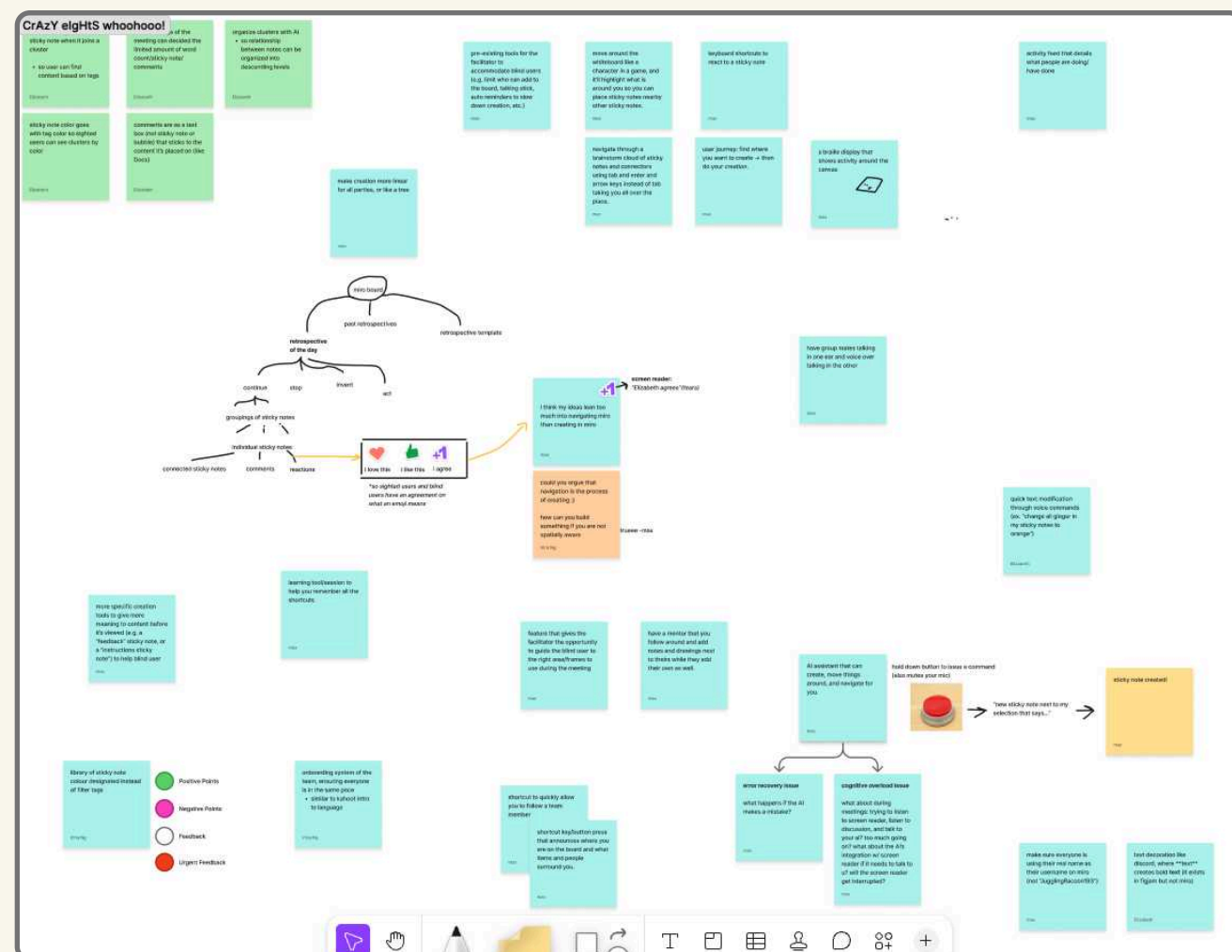
With tools that match the task you're working on

First-time users

Since most blind users aren't going to be familiar with whiteboarding

We did a Crazy 8s brainstorming session.

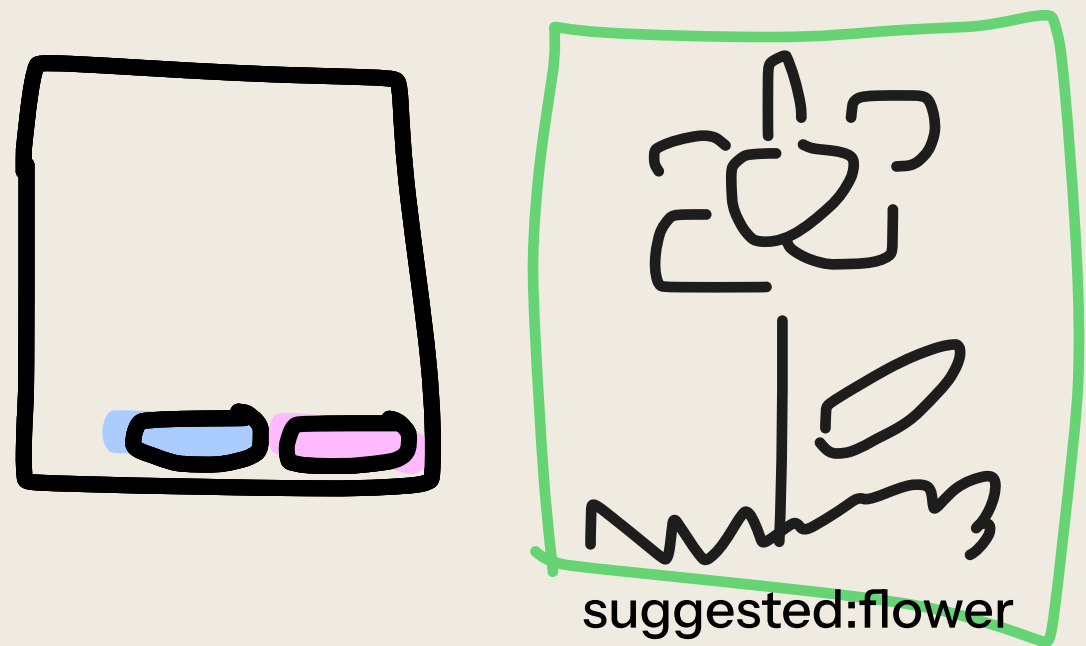
Then, we selected and expanded on our **three strongest ideas** — chosen based on quality of evidence, realism, and potential for positive impact.



see next page →

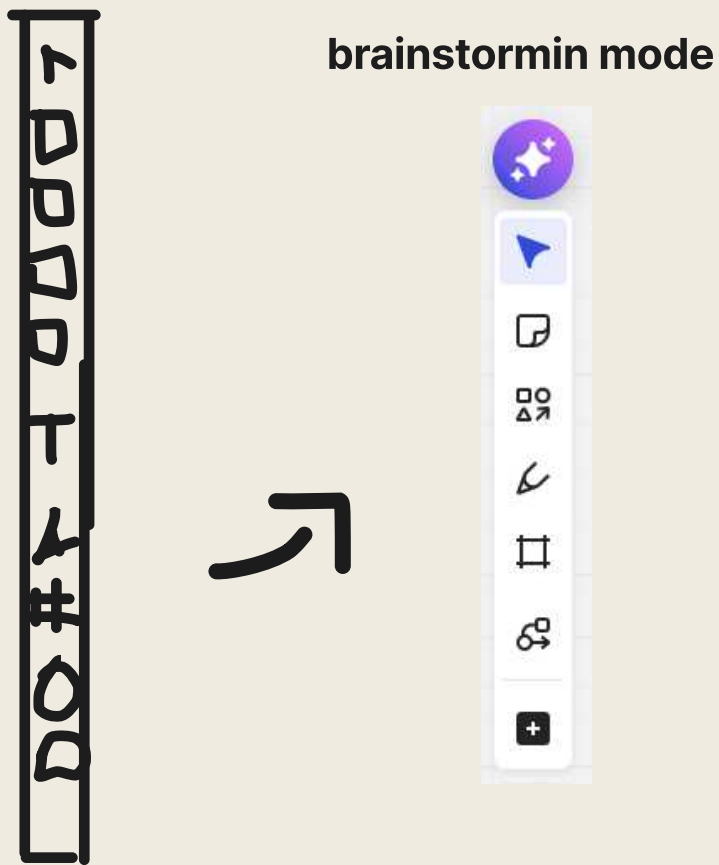
Out of our strongest ideas...

Automatic labelling tagging and grouping



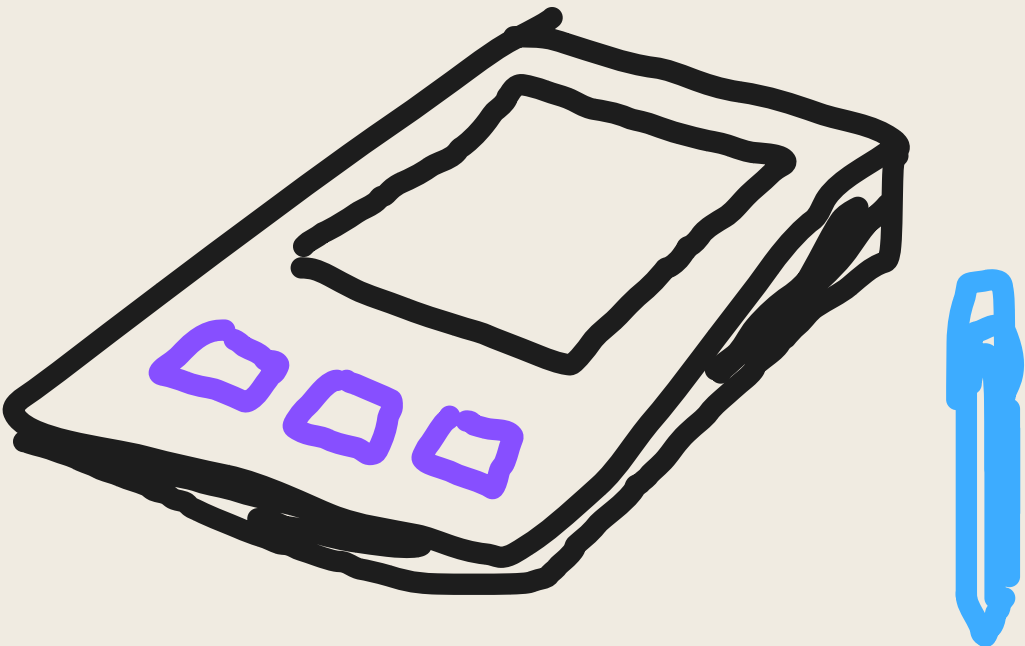
Adding captions and alt text for sighted users *while* items are created (not just at the end)

Customizable toolbar



Limiting the toolbar to whichever tools are chosen by that particular user

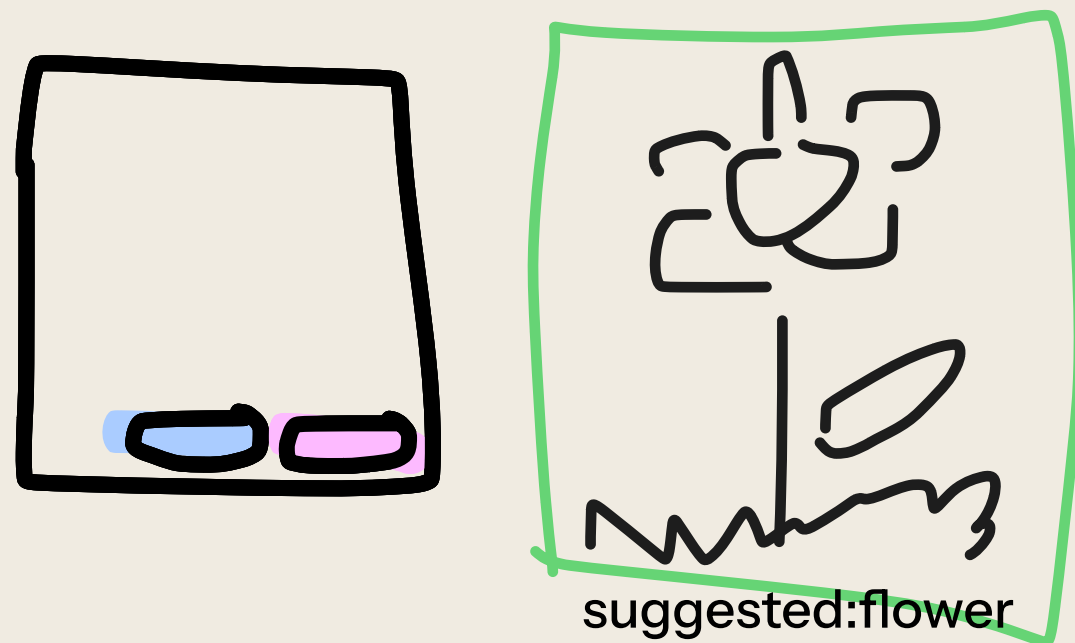
External drawing tablet



Giving a spatial representation of the whiteboard with custom shortcuts for key actions

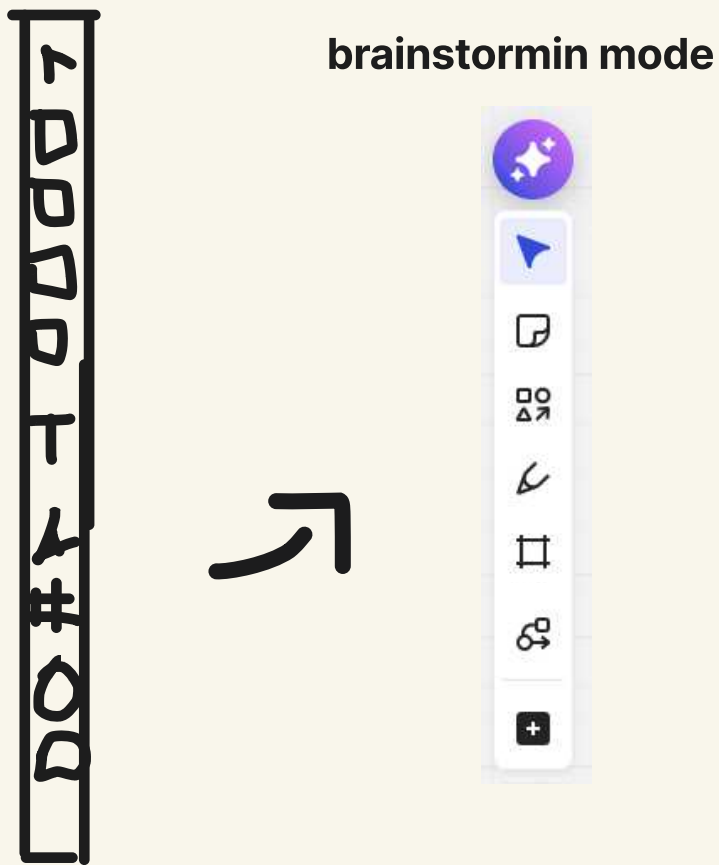
... we chose **this one.**

Automatic labelling tagging
and grouping



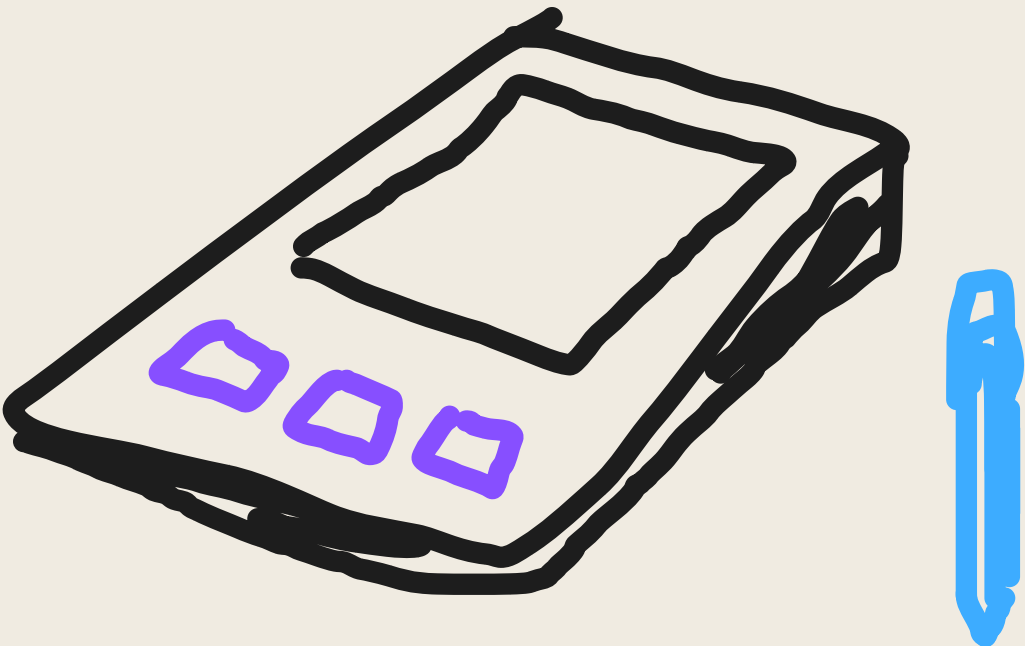
✗ Focused more on navigation
rather than creation (strays
from the brief)

Customizable toolbar

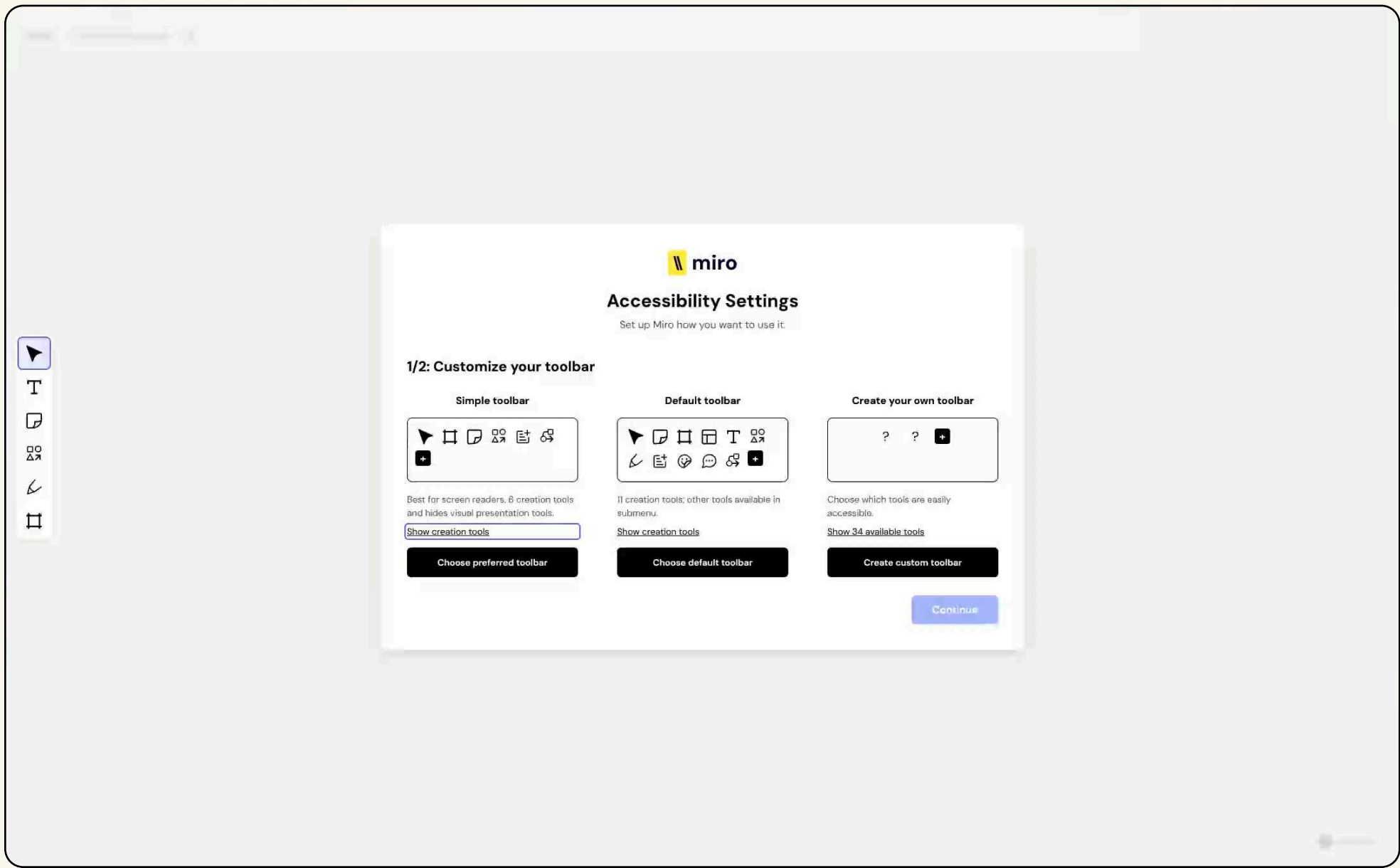


✓ Aligns with our insights
✓ Can be expanded into more
accessibility features later

External drawing tablet



✗ Less realistic for *Miro* to
implement
✗ Users unlikely to spend money
on this *one particular tool*



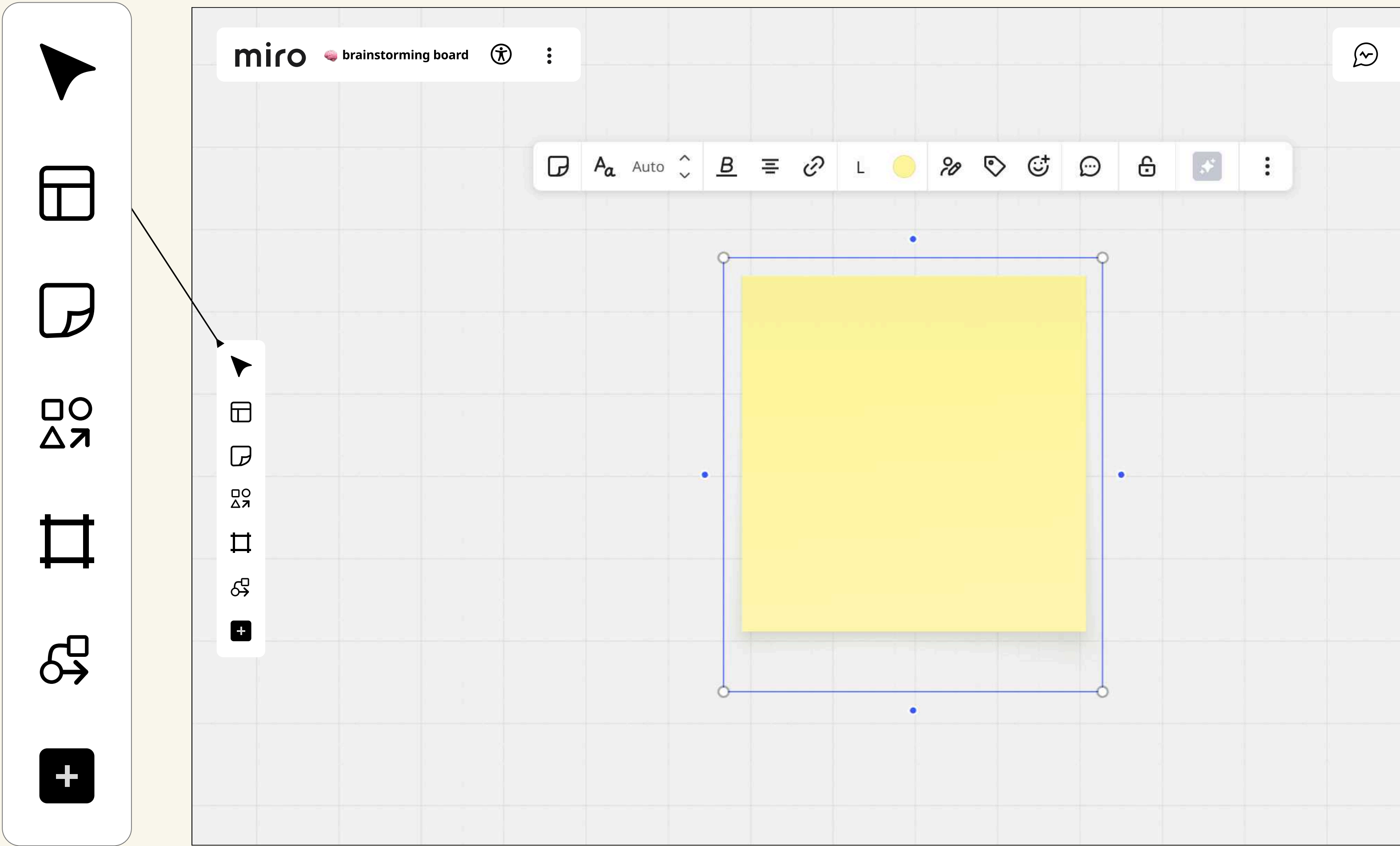
Customizable toolbar modes

Enabling choice and control over your preferred whiteboard features, streamlining the steps to find the tool you need

Customizable Toolbars

Customization & control over what your default actions you have available on the toolbars in Miro.

Removes unnecessary tools that blind users might not often use



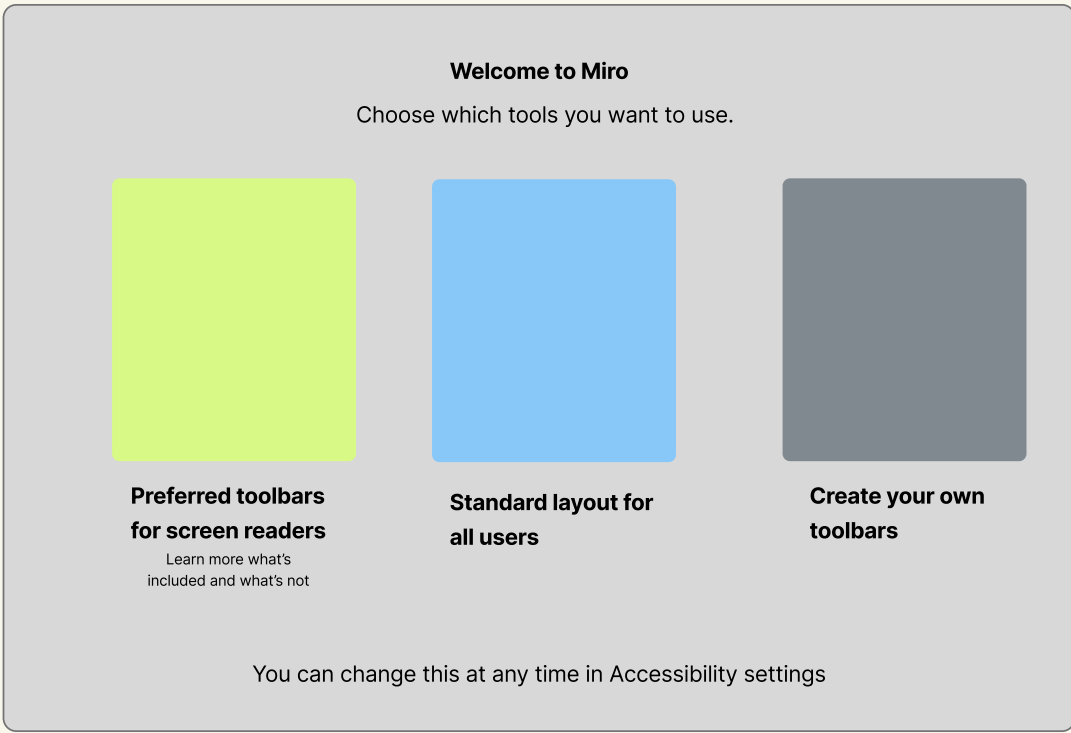
Simple Toolbar

Welcome and setup

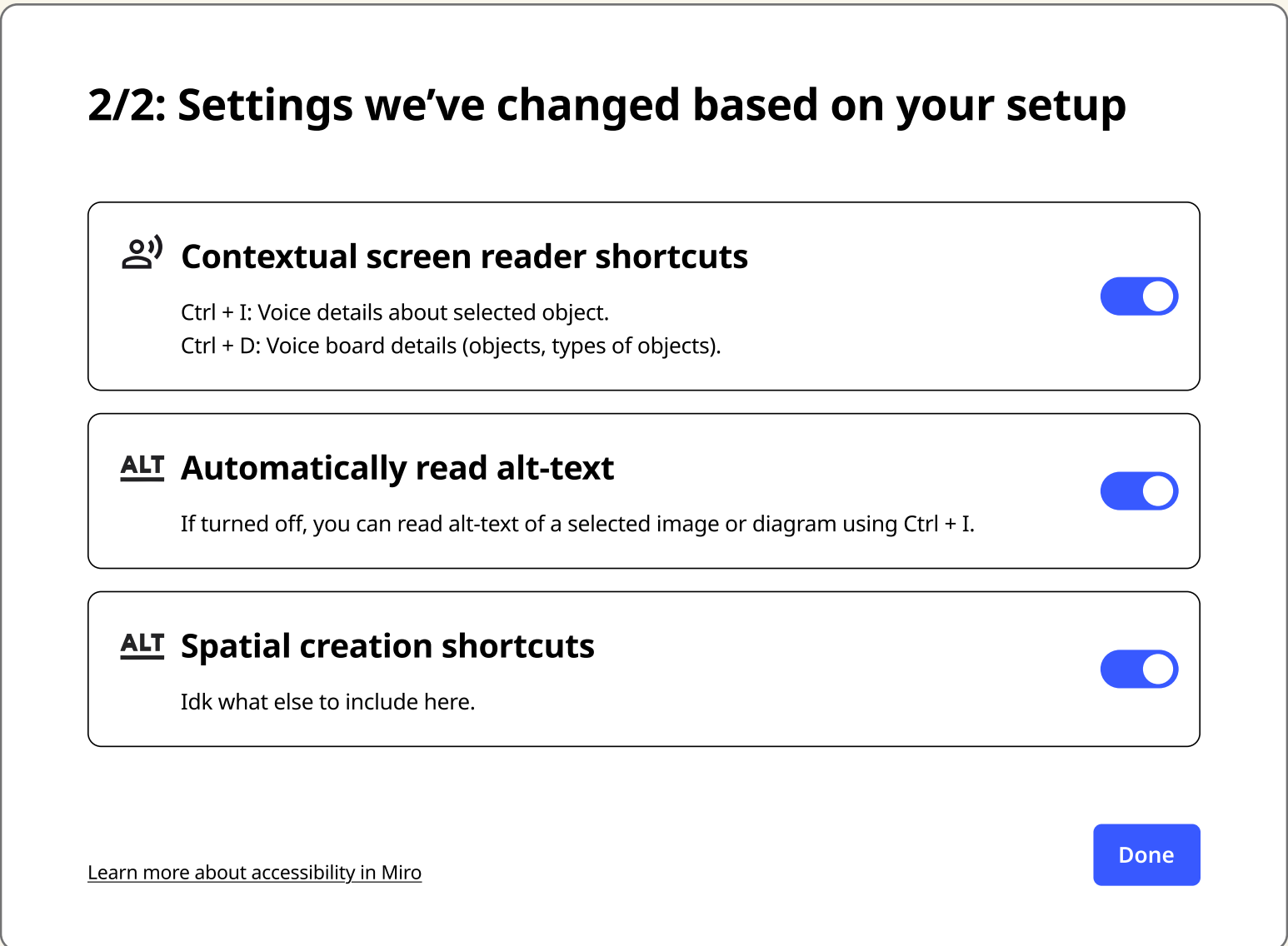
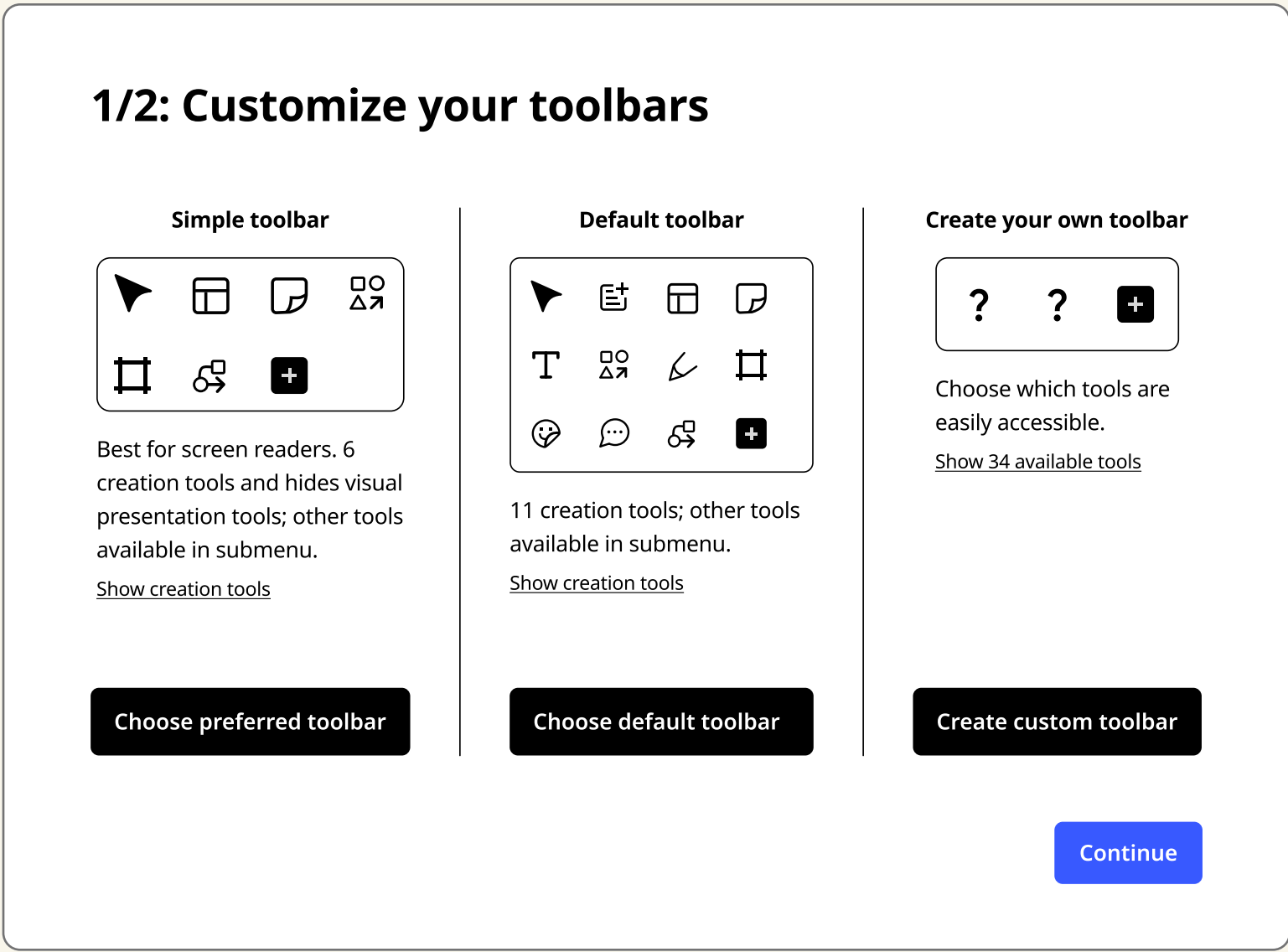
When opening their first board, Miro prompts the user to choose their toolbar settings.

Choosing the “Simple toolbar” also enables Miro’s other existing accessibility settings.

Wireframe



Mockups (Figma)



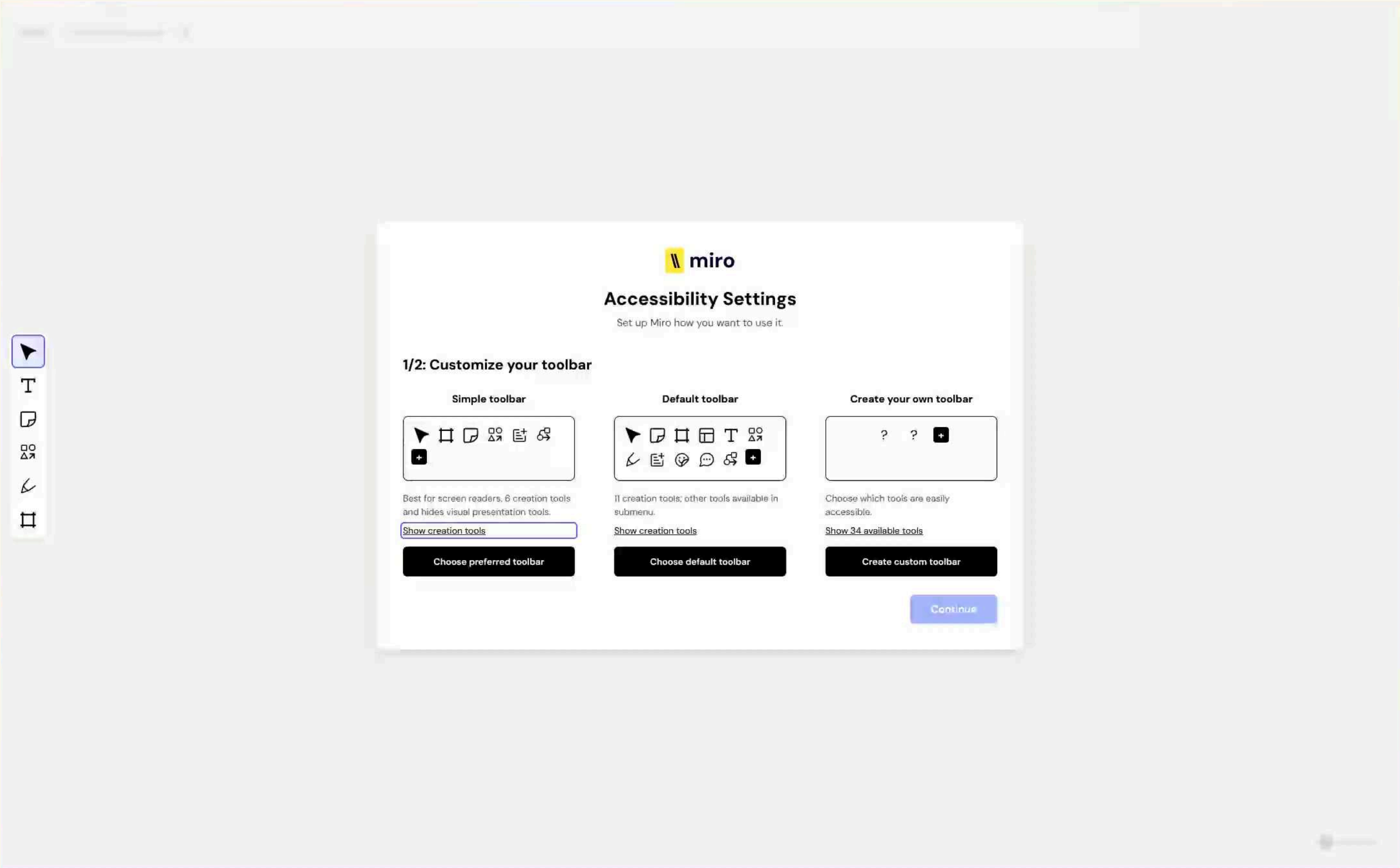
Prototype (assisted by Gemini, Claude Code)

Welcome and setup

On a screen reader...

Right now, entering the Miro board immediately prompts you to jump to the first object, without any context of what’s there.

Instead, having a dedicated setup section can set expectations and feel more guided.

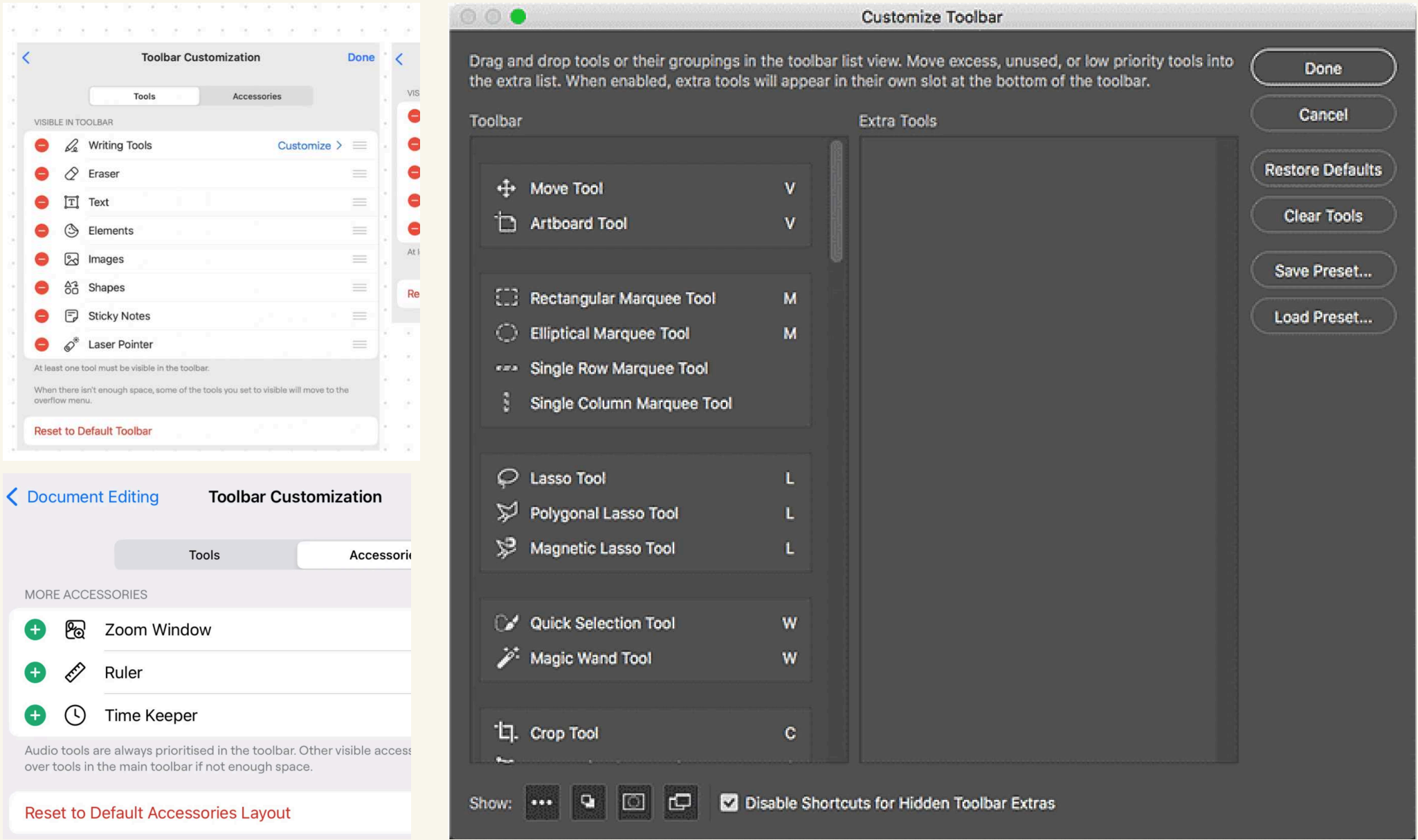


Final concept

Toolbar customization menu

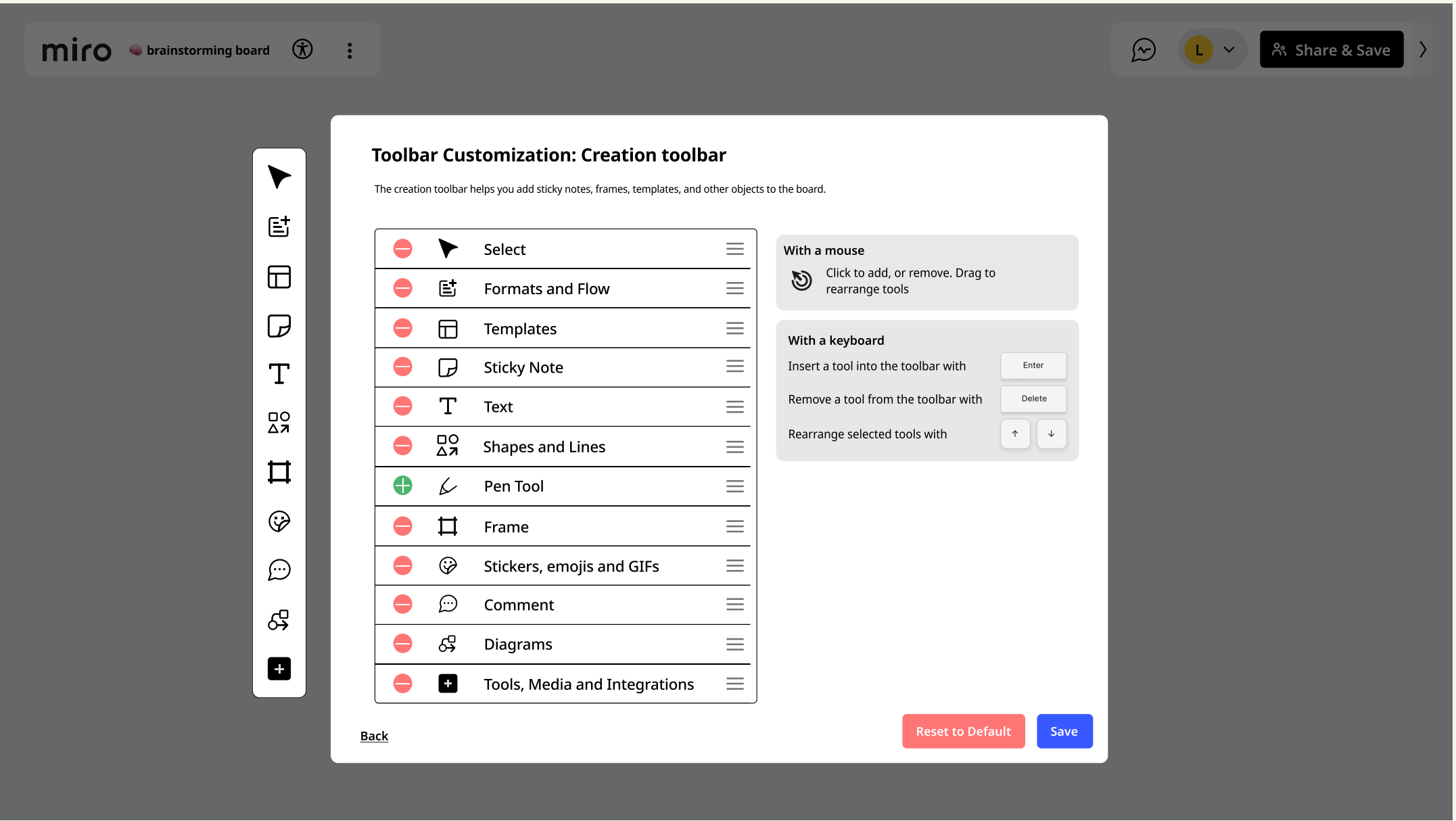
Using the keyboard or mouse, the user can shift, add and delete the whiteboard tools to cater to their needs.

Inspiration



Screenshots from GoodNotes, Adobe Illustrator

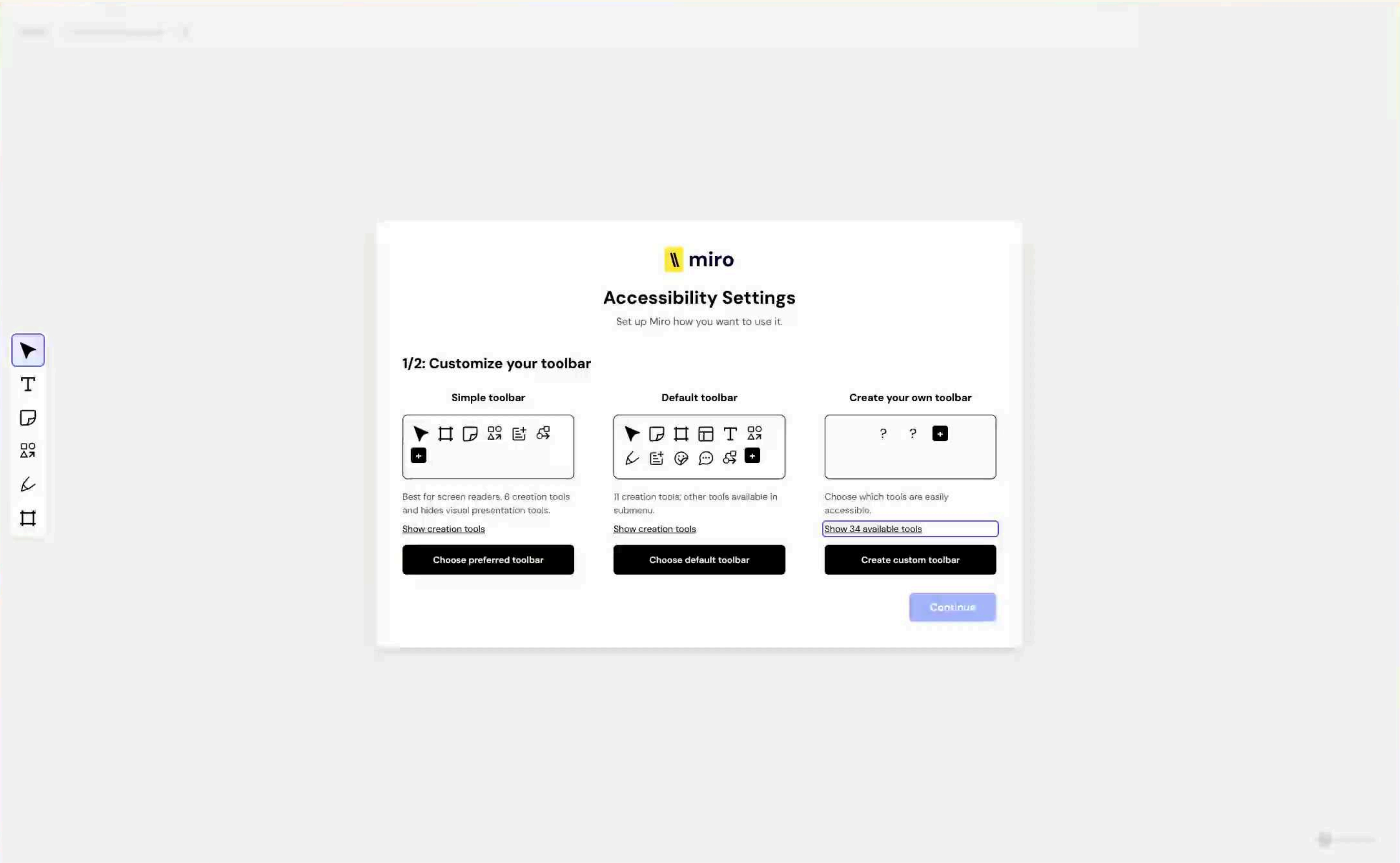
Mockup (Figma)



Toolbar customization menu

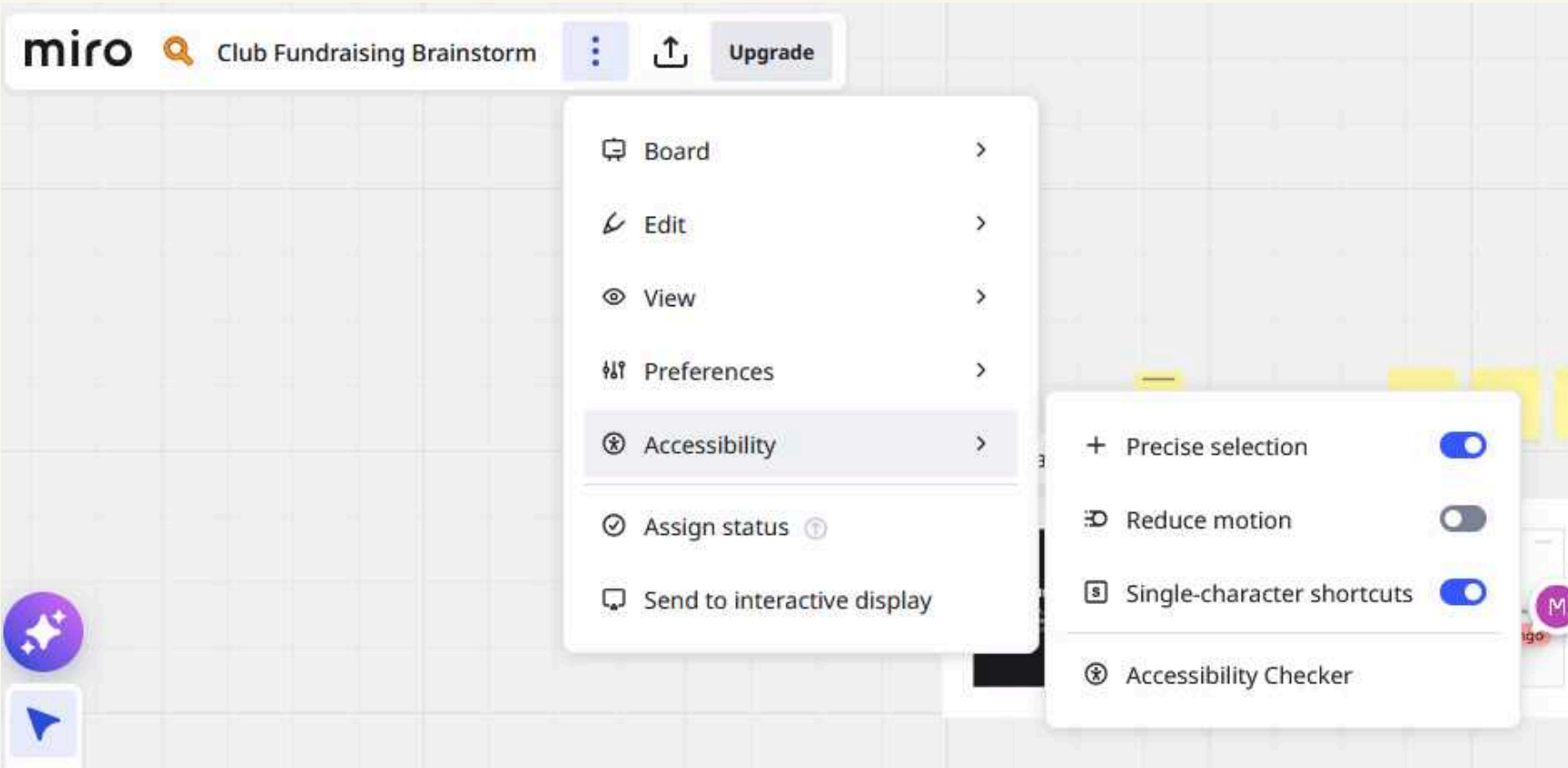
Using the keyboard or mouse, the user can shift, add and delete the whiteboard tools to cater to their needs.

Prototype (assisted by Gemini, Claude Code)



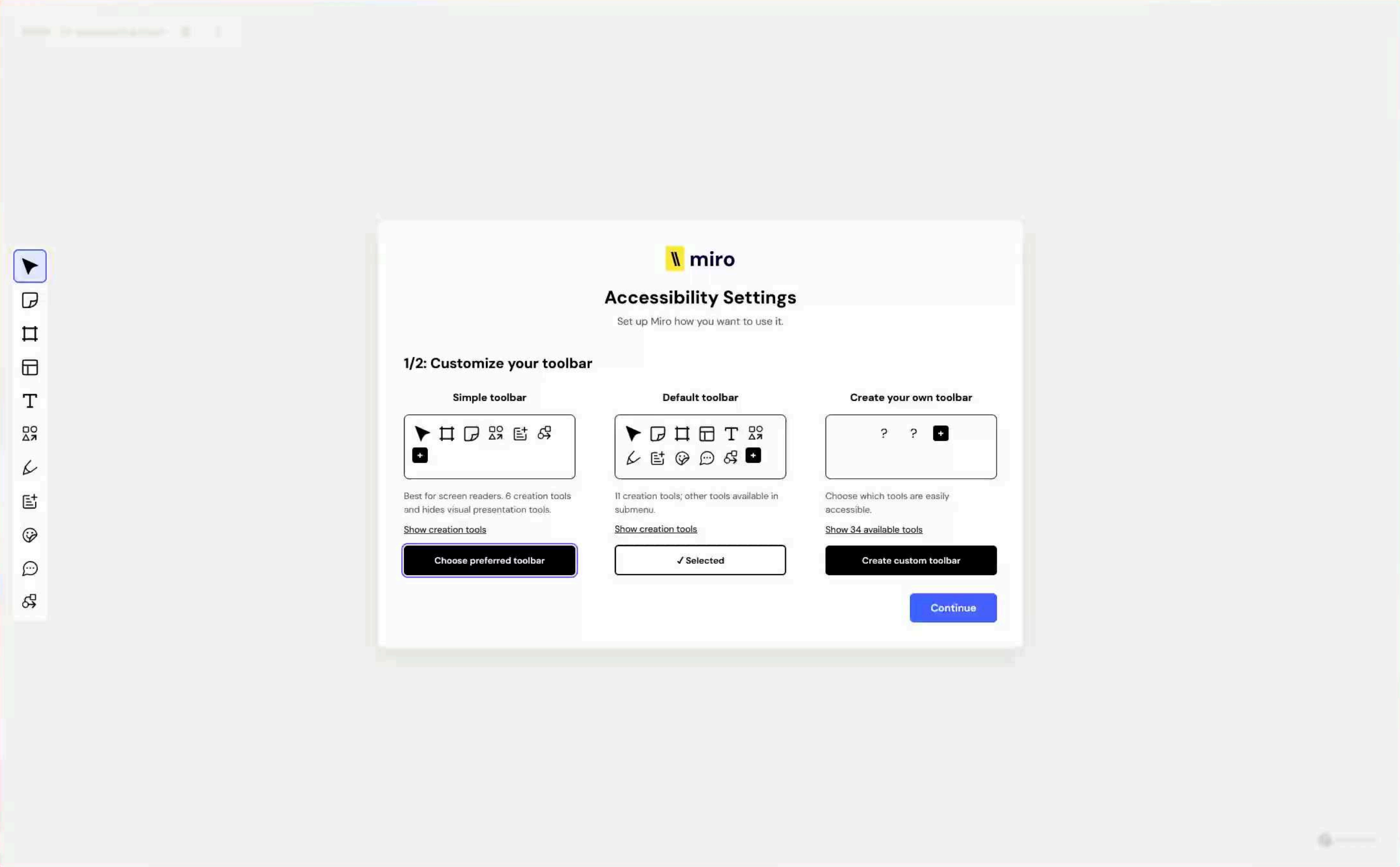
Quick access to accessibility settings

As a screen reader user, you should be able to quickly modify your settings without digging through a submenu.



Miro’s current accessibility menu

Prototype (assisted by Gemini, Claude Code)



Design Goals to Screen Reader Modes

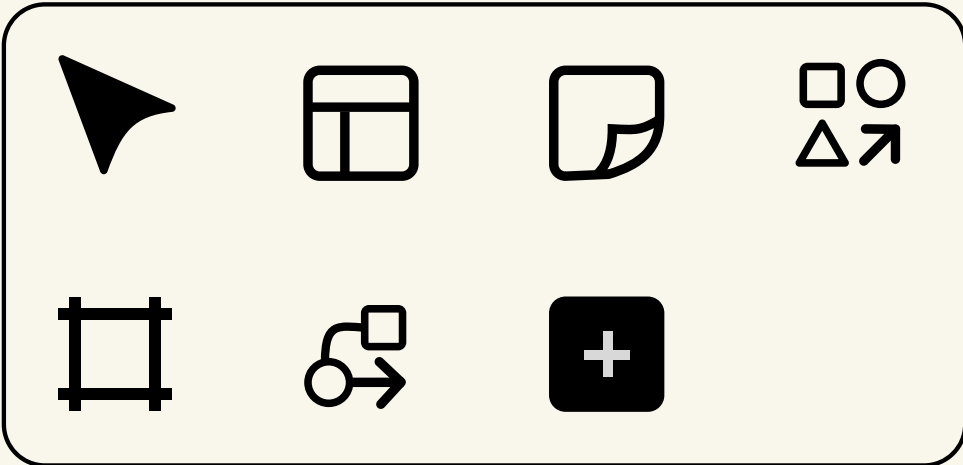
Enable flexible customization of the program to accommodate the needs of a blind user

Reduce functional clutter so they can focus on creating with their team.

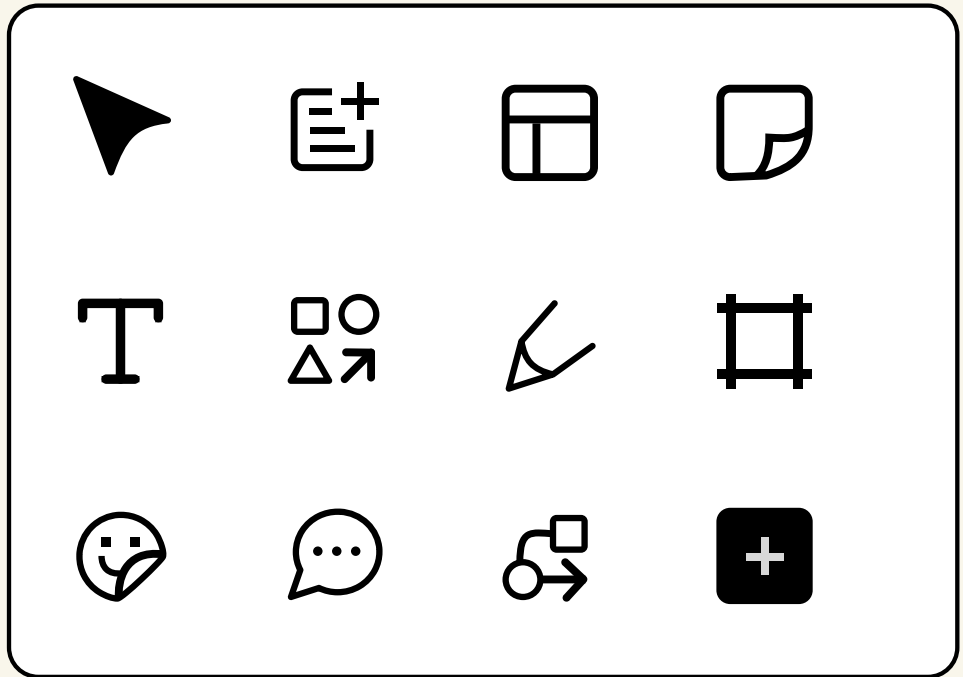
“ Being able to choose the information felt really important and meaningful...it would save time not having to listen to information I didn’t need or care about.

-Research Study Blind Participant

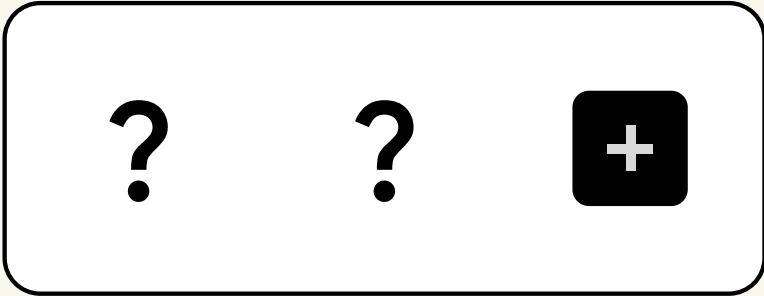
Simple toolbar



Default toolbar

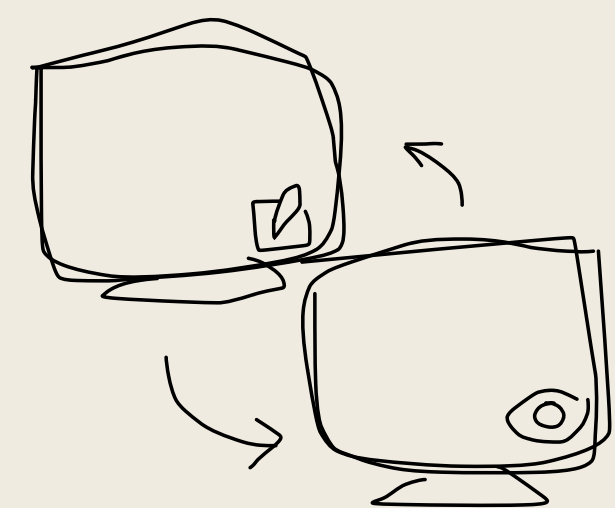


Create your own toolbar



Our next step would be...

Viewing mode and Creating Mode



Toggle between exploration and creation, having focused display for each mode

Flushed out Prototype



Pushing all the design elements. Fixing all the missing bugs and glitches

Claude Code & Gemini were used to create the prototype

These LLMs were used to convert our Figma mockups into a working static webpage. We took screenshots of our designs and attached a description detailing what we wanted to create.

You can view our conversations with AI [here](#).

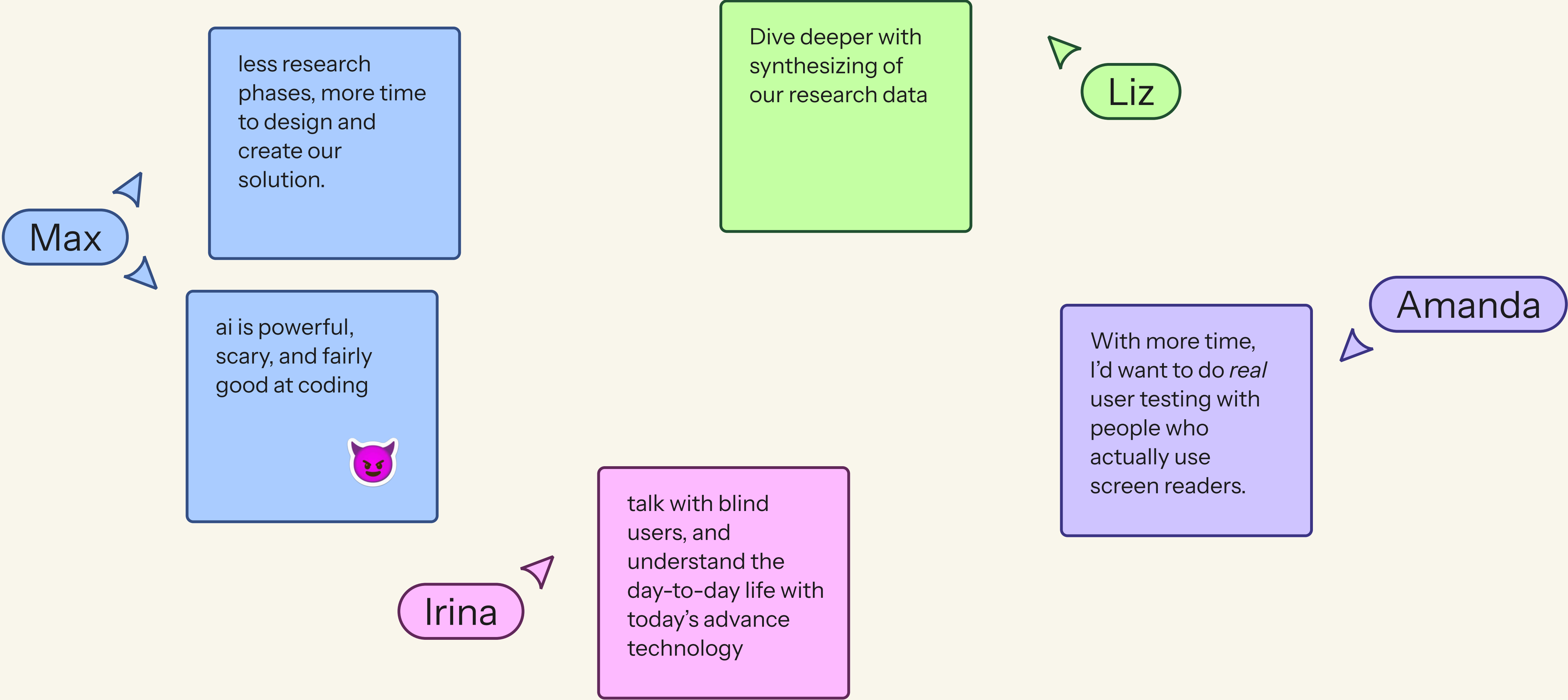
Mentor feedback:

When asked how to make our prototype, since you can't keyboard-navigate Figma: "Just vibe code it."

Figma's AI was used to edit images for our video submission

It removed backgrounds, enhanced the resolution, and removed objects.

A few lessons learned...



Reference

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